hologram. Even if the hologram is broken into pieces, single pieces produce the whole image of the body with reduced intensity.

Holography is used for diagnosis in various fields of medicine, non-destructive testing, holographic information storage, display devices and pattern-matching procedures in credit cards and identity card verification. It is also used for establishing a secret communication system by storing secret documents, maps and objects as holograms and reconstructing the image only at the receivers' end. It is expected that in the near future, holography may even be used for target recognition from air to ground, and we may have holographic movies and TVs.

Communication. Properties of a laser like wide bandwidth and narrow beam width over long distances enable its utility in this field. The semiconductor laser is generally used for optical-fibre communication, which is excited directly by electric current to yield a laser beam in the infrared region. Capacity of the communication channel is directly proportional to the bandwidth of frequency. So at optical frequencies, the information-carrying capacity is higher than at lower frequencies.

Communication of signals where light is used as a signal carrier and optical fibres as transmission medium is called optical-fibre communication. The first ever optical-fibre communication system was established in 1977. Since then, millions of dollars have been spent on long-distance communication where data or signals are converted into light pulses or codes using a suitable light source.

Light signals are transmitted through the core of optical fibres, amplified at the receiving end and converted into readable electrical signals by decoding light signals and, hence, getting the required original information. As light has high information-carrying capacity, optical-fibre communication is most probable these days. Moreover, light can easily be transmitted through extremely-thin hair-like fibres to large distances without reducing the intensity.

Advantages of optical-fibre communication are:

1. High information-carrying capacity
2. Free from electromagnetic interference
3. Lightweight
4. Minimum signal leakage

Due to these advantages this system finds many applications in the field of telecommunications. The telecommunication market for optical fibres has exploded in several developed countries like the USA, the UK, France, Denmark, Germany and Japan.

Medicine. No talk on laser in medicine can be done without mentioning Leon Goldman, the father of laser medicine. He was the first to use laser to treat a skin disease, which developed as dermatology.

Photo-rejuvenation is a process in which lasers are used to evaporate moisture from tissues responsible for wrinkles, dark spots and the like, from the face and create a layer of self-healing wounds.

Lasers have been extensively used in surgery, the very first being eye surgery where a laser was used to weld detached retinas and photocoagulate the vessels that grow in retinas,

Institutes/Research organisations in the field of lasers

1. RRI (DST), Bengaluru, Karnataka
2. IISc, Bengaluru, Karnataka
3. LEOS (ISRO), Bengaluru, Karnataka
4. Manipal University, Karnataka
5. NCBS, Bengaluru, Karnataka
6. PRL, Ahmedabad (ISRO), Gujarat
7. SGSIT, Indore, Madhya Pradesh
8. DAVV, Indore, Madhya Pradesh
9. RRCAT, Indore (DAE), Madhya Pradesh
10. UGC-DAE CSR - Indore, Madhya Pradesh
11. TIFR, Mumbai (DAE), Maharashtra
12. BARC (DAE), Mumbai, Maharashtra
13. Pune University, Maharashtra
14. SAMEER, Mumbai (MeitY), Maharashtra
15. DIAT, Pune (DRDO), Maharashtra
16. IIT Mumbai, Maharashtra
17. UGC-DAE CSR - Mumbai, Maharashtra
18. IRDE, DEHRADUN (DRDO), Uttarakhand
19. ARCHEM, University of Hyderabad, Telangana
20. CLPM, ARCI - Hyderabad (DST), Telangana
21. IIT, Patna, Bihar
22. LASTEC (DRDO), New Delhi
23. IIT Delhi, New Delhi
24. Delhi University, Delhi
25. NPL, Delhi (CSIR), Delhi
26. IISER, Mohali, Punjab
27. NCESS (MoEs), Kerala
28. ISP, CUSAT (Kochi) - Centre of Excellence in Lasers & Optoelectronic Sciences (CELOS), Kerala
29. IIST (ISRO), Trivendrum
30. Sathyabama University, Chennai, Tamil Nadu
31. IGCAR, Kalpakkam (DAE), Tamil Nadu
32. NCUFP, University of Madras, Tamil Nadu
33. IIT Madras, Tamil Nadu
34. UGC-DAE CSR - Kalpakkam, Tamil Nadu
35. SN Bose National Centre For Basic Sciences, Kolkata (DST), West Bengal
36. IICB (CSIR), Kolkata, West Bengal
37. Calcutta University, Kolkata, West Bengal
38. Jadavpur University, Jadavpur, West Bengal
39. IISER, Kolkata, West Bengal
40. SINP (DAE), Kolkata, West Bengal
41. IIT Kharagpur, West Bengal
42. IACS (DST), Kolkata, West Bengal
43. UGC-DAE CSR, Kolkata, West Bengal
44. IIT Kanpur, Uttar Pradesh
45. IIT Guwahati, Assam
46. BS Abdur Rahman University, Chennai, Tamil Nadu
47. University of Kerala, Thiruvananthapuram, Kerala